

# Equity Selector

Research Pipeline and Strategy Validation Methodology

Project Documentation

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# 1 Executive summary

Equity Selector is a multi-stage quantitative equity research system. Its intended end product is a repeatable portfolio allocation process built from screened market data, target-specific predictive models, model-quality information, horizon-aware signal weights and portfolio-level evaluation.

The project is designed to identify promising equity strategy candidates, explain how those strategies behave and carry a frozen set of research decisions into a later evaluation period. It connects statistical prediction with the practical construction of a long-only equity portfolio.

The research begins with High Liquidity 30 and uses daily adjusted OHLCV observations, with the S&P 500 as the market reference. The account profile is a GBP-based Trading 212 Invest portfolio with £4,000 initial capital, individual USD trades and retained USD cash. Portfolio research includes a 20% GBP drawdown acceptance threshold.

The project separates five questions:

- **Data suitability:** which stocks and feature columns form a suitable research panel before target performance is considered?
- **Target predictability:** which features and model configurations remain competitive through chronological validation?
- **Model confirmation:** how does a selected model perform on a later model-test block?
- **Portfolio usefulness:** how should target meaning, model quality, prediction horizon and allocation settings turn predictions into portfolio weights?
- **Strategy behaviour:** how do returns, risk, costs, concentration and parameter sensitivity explain a complete strategy?

## 2 Research philosophy

### 2.1 Chronological split

Each period has a defined research role. Earlier observations fit models; later development observations validate model choices; a subsequent block confirms selected models. Horizon research, portfolio selection and final strategy evaluation then use their respective chronological windows. Target purging is applied at supervised learning boundaries.

## **2.2 Target-blind screening before target selection**

The first screens ask whether a stock or feature is structurally suitable. They use training observations to assess coverage, validity, continuity and the information structure of the feature panel. Target-specific analysis follows after this common research pool has been formed.

## **2.3 Model accuracy and portfolio usefulness**

The project records statistical performance separately from economic purpose. A model supplies evidence about one target; the portfolio layer decides how that evidence contributes to an allocation. This distinction supports the combination of ranking, direction, risk, opportunity and special-purpose signals.

- Statistical target type and model parameters identify the prediction task.
- Fold-level performance and variability describe the model through time.
- Predictability and Quality Scores summarise the selected target model.
- Portfolio Target Type and Horizon identify the signal's economic role.
- Horizon Score and Type Score determine its influence in portfolio construction.

## **2.4 Search efficiency and progressive selection**

Research work is organised around reusable inputs and explicit selection rules. Target-specific feature loading keeps each model task focused. Progressive pruning removes less competitive configurations at scheduled folds. Prediction caches allow horizon and portfolio settings to be compared without repeatedly fitting the same selected model.

The intended model-research configuration uses 63-date validation windows, pruning at folds 3, 5, 7, 9, 11 and 14, and a minimum eligibility count of 15 folds. Models that remain active continue through the scheduled validation calendar. The minimum eligibility count is a qualification rule, while the pruning schedule determines when configurations are compared and removed.

## **3 End-to-end architecture**

The project is organised into three research phases: model research, portfolio research and final evaluation. Original root scripts expose the settings for each stage, while focused Python packages supply the shared calculations, data handling, validation and scoring functions.

### 3.1 Research pipeline and hand-offs

```
Prepare Research.py [model phase]
|
Data_Creation_Screening.py
+-- Features_Targets_Data.db
+-- Selected_Features.txt
|
Model_Fitting.py
+-- Validation_Model_Fits/High_Liquidity_30.db
|
Best_Model_Test.py
+-- Final_Test_Results.db
|
Horizon Score Backtests.py
+-- Top_Horizon_Scores.txt
|
Prepare Research.py [selection phase]
|
Backtest Database.py [selection cache]
+-- Backtest_Database.db
|
Backtest Simulations.py
+-- Portfolio_Simulation_Results.db
+-- Passed Strategies in the selection cache
|
Precise Backtest.py
+-- Detailed strategy evaluation and daily returns
|
GBP Portfolio Check.py
+-- GBP_Check/GBP_Selection.db
|
Prepare Research.py [final phase]
|
Backtest Database.py [final cache]
|
Frozen Final Test.py
+-- Frozen_Final_Evaluation.db
+-- Summary, daily returns and research manifest
```

Run Research.py launches one selected stage and applies the numerical-library thread settings before importing it. Prepare Research.py creates the named phase directories and stages the completed inputs required by the next phase. Intraday Conversion.py provides a separate session-aware preparation route for intraday tables.

### 3.2 Script roles

| Script                     | Research role                                           | Principal output                               |
|----------------------------|---------------------------------------------------------|------------------------------------------------|
| Data_Creation_Screening.py | Build and screen the daily research panel.              | Research table and selected-feature mapping.   |
| Model Fitting.py           | Compare model configurations in walk-forward folds.     | Target-level validation databases.             |
| Best_Model_Test.py         | Confirm selected models and prepare portfolio metadata. | Final_Test_Results.db.                         |
| Horizon Score Backtests.py | Compare the importance of target/horizon components.    | Retained horizon-score configurations.         |
| Backtest Database.py       | Fit selected models and cache later predictions.        | Backtest_Database.db.                          |
| Backtest Simulations.py    | Compare full portfolio settings and robustness.         | Simulation results and selected strategies.    |
| Precise Backtest.py        | Reconstruct and examine selected configurations.        | Detailed evaluation tables and return streams. |
| GBP Portfolio Check.py     | Evaluate the GBP account and drawdown criterion.        | GBP-approved strategy selection.               |
| Frozen Final Test.py       | Evaluate every frozen finalist on the final cache.      | Final-period summary and daily results.        |

## 4 Quantitative definitions

### 4.1 Feature and target reference

The companion Features & Targets Quant Reference provides the mathematical definitions used in the feature and target libraries. Each family describes the quantities being measured, the required market observations, the observation window and the resulting output columns.

Features describe information available at the observation date. Targets describe the future outcome the model is trained to estimate. The target horizon therefore also determines the separation required at model-fitting boundaries.

## 4.2 Feature families used in data creation

Feature construction follows a dependency order. The individual-stock library is applied before calculations that require the benchmark or a cross-stock panel.

| Family group                      | Quantities represented                                                            |
|-----------------------------------|-----------------------------------------------------------------------------------|
| Returns and momentum              | Past simple/log returns, cumulative changes, momentum and reversal measures.      |
| Volatility and range              | Realised volatility, downside/upside variability and OHLC range measures.         |
| Trend and moving averages         | Trend strength, moving-average relationships and distance from smoothed prices.   |
| Drawdown and tails                | Historical drawdowns, adverse return tails and distributional characteristics.    |
| Volume and liquidity              | Volume behaviour, dollar-volume relationships and liquidity-related measurements. |
| Market-relative features          | Relative returns, beta, correlation and residual stock behaviour.                 |
| Technical and regime features     | Technical transformations and descriptions of prevailing market conditions.       |
| Interactions and composites       | Combined feature relationships and cross-family summaries.                        |
| Cross-sectional and market panels | Ranks, breadth, dispersion and comparisons across the available stock universe.   |

## 4.3 Target families used in data creation

The target library covers forward returns, realised future volatility, direction and event labels, barriers, favourable and adverse excursions, drawdown paths, risk-adjusted outcomes and cross-sectional future ranks.

- **Continuous targets** express numerical future outcomes or relative rankings.
- **Binary targets** represent two-state outcomes, including directional or threshold events.
- **Multiclass targets** distinguish several future states or categories.

The same target can have several horizons. Each horizon is retained as explicit portfolio metadata so later stages can compare signals operating over different time scales.

## 5 Stage I - Data\_Creation\_Screening.py

### 5.1 Research universes

The initial research universe is High Liquidity 30. The script also holds explicit comparison universes that can be selected for separate research runs.

| Stock type                 | Size | Research composition                             |
|----------------------------|------|--------------------------------------------------|
| High Liquidity 30          | 30   | High-liquidity equity selection.                 |
| Medium Liquidity 30        | 30   | Medium-liquidity comparison group.               |
| Lower Liquidity 30         | 30   | Lower-liquidity comparison group.                |
| Sector Spread 30           | 30   | Sector-spread selection across liquidity groups. |
| Liquidity Barbell 30       | 30   | High- and lower-liquidity components.            |
| Institutional Liquidity 60 | 60   | High- and medium-liquidity groups.               |
| Medium Small Liquidity 60  | 60   | Medium- and lower-liquidity groups.              |
| Medium Large Liquidity 60  | 60   | Large/medium-liquidity composition.              |
| All Liquidity 90           | 90   | Combined liquidity spectrum.                     |

### 5.2 Source history and benchmark

Daily research draws adjusted OHLCV data from Yahoo Finance. The chosen ticker set and the market reference, ^GSPC, use the configured download dates. The research download begins in 1997 to supply history ahead of the first modelling period and extends to the end of the horizon-research data window.

OHLCV observations support price, volatility, path and liquidity-related features. The benchmark provides the common market series used for relative features and strategy comparisons. Each row is identified by Date and Ticker.

### 5.3 Stock screening before feature engineering

Stock-level screening is applied to training observations before the main feature panel is constructed. The sequence checks sample size, missingness, invalid observations and continuity relative to the market calendar. Securities passing the selected screen become the feature-engineering universe.

## 5.4 Feature creation and target-blind screening

The feature builder creates individual-stock measures, market-relative quantities and cross-sectional information. The target-blind screen then examines missingness, invalid values, variance, duplicate construction and feature-to-feature correlation. This produces a common pool of features for subsequent target-specific analysis.

## 5.5 Target creation and chronological split

Target columns are constructed from the price panel and placed alongside the features. The explicit research boundaries divide observations into training, validation and model-confirmation periods. Target-specific selection is performed on the training partition.

## 5.6 Target-specific feature screens

| Target type | Screening sequence                                                                                                                  |
|-------------|-------------------------------------------------------------------------------------------------------------------------------------|
| All types   | Feature-target coverage and time stability.                                                                                         |
| Continuous  | Quantile spread, quantile monotonicity, Pearson correlation and cross-sectional information-coefficient screening where applicable. |
| Binary      | Quantile spread, quantile monotonicity and Pearson correlation.                                                                     |
| Multiclass  | Coverage and time-stability analysis.                                                                                               |

Each target begins with its own copy of the common feature pool. The resulting mapping records the retained feature names for that target under the selected stock-universe key in `Selected_Features.txt`. Downstream stages load the mapping by universe and target.

## 5.7 Stored research table

`Features_Targets_Data.db` stores the stock-type research table. Its columns contain date/ticker identity, the retained base data, target columns and the union of selected features. Model fitting can then request only the columns relevant to its current target.

## 5.8 Intraday session conversion

`Intraday_Conversion.py` applies session-aware feature lookbacks and target horizons to intraday tables. It groups observations by ticker and session, identifies the required start-of-session history and reserves the forward rows needed by each target. Explicit override dictionaries describe custom column requirements. The initial daily research sequence proceeds directly to model fitting.

## 6 Stage II - Model Fitting.py

### 6.1 Target-specific loading

The model-fitting stage reads the selected universe schema and processes one target at a time. The working frame contains Date, Ticker, the target and its selected features. Target/feature cleaning establishes the observations available for that task. Statistical target type determines the estimator families and evaluation metrics.

### 6.2 Active model catalogue

The extensive research profile uses the grid catalogue. Model families span baseline predictors, linear and regularised models, robust regression, tree ensembles, boosting, support-vector methods, neural networks and nearest-neighbour methods. Classification tasks use the corresponding supported classification families.

| Setting group           | Research values                                                        |
|-------------------------|------------------------------------------------------------------------|
| Ridge alpha             | 0.01, 0.1, 1, 10, 100                                                  |
| Sparse-model alpha      | 0.00001, 0.0001, 0.001, 0.01                                           |
| Classification C        | 0.1, 1, 10                                                             |
| L1 ratio                | 0.25, 0.75                                                             |
| Boosting learning rates | 0.03, 0.1                                                              |
| Class weights           | Unweighted and balanced                                                |
| Random forest           | 300 trees; depths 3 and 6; leaf sizes 20 and 50                        |
| XGBoost                 | 300 estimators; depths 2 and 4; regularised, subsampled configurations |
| LightGBM                | 300 estimators; 7 and 15 leaves; minimum child size 50                 |
| MLP and KNN             | 32-unit hidden layer; 20, 50 and 100 neighbours respectively           |

### 6.3 Configuration history

A model configuration is identified by its estimator family and parameter dictionary. Each target stores fold-level observations and a search summary in its universe-specific validation database. The history supports the identification of previously evaluated configurations and the tracing of the metrics used in selection.

The principal table names are <target>\_\_folds for individual fold results and <target>\_\_search for the aggregated search leaderboard. Target identity, model parameters and fold information connect the stored observations.

## 6.4 Chronological walk-forward folds

The validation calendar is divided into consecutive windows of 63 observed dates. For each fold, the model is fitted on observations before the validation window. The training boundary is purged by the target's forward horizon. Any fitted preprocessing is learned from that training sample before being applied to the later validation observations.

The 2007-2012 validation period provides approximately 24 folds when the full daily calendar is represented. The final window uses the remaining observed dates. Each target follows its own available validation calendar after its feature and target requirements are applied.

## 6.5 Progressive pruning schedule

The intended research run retains the original progressive schedule. At each listed fold, the configured fraction and maximum count determine the retained comparison set. The portfolio role and statistical target type guide the metrics used to compare configurations.

| Completed fold | Retained fraction | Maximum configurations |
|----------------|-------------------|------------------------|
| 3              | 95%               | 20,000                 |
| 5              | 90%               | 5,000                  |
| 7              | 80%               | 1,000                  |
| 9              | 65%               | 300                    |
| 11             | 50%               | 70                     |
| 14             | 50%               | 15                     |

## 6.6 Continuing validation and eligibility

Pruning determines which models remain active; it does not define the end of the validation calendar. A model that survives the fold-14 stage continues through the remaining folds. The research minimum of 15 validation folds is an eligibility condition used with the other performance and stability gates.

For each active model, the accumulated results describe both average prediction quality and variation between folds. Model selection therefore considers performance through time rather than a single successful interval.

## 6.7 Leaderboard metrics

Continuous tasks record RMSE, normalised RMSE, MAE,  $R^2$  and Rank IC. Binary tasks record ROC AUC, PR AUC, log loss and F1-related measures. Multiclass tasks record Macro F1, balanced accuracy and log loss. The leaderboard combines the relevant fold statistics with model identity and portfolio-role information.

## 6.8 Testing Eligible gates

The validation leaderboard identifies candidates suitable for the model-confirmation stage. Eligibility combines the required fold count with the applicable performance and variability rules. The stored eligibility reason explains how the decision was reached.

| Stability setting                              | Research value     |
|------------------------------------------------|--------------------|
| Maximum continuous Rank IC standard deviation  | 0.10               |
| Maximum binary ROC AUC standard deviation      | 0.07               |
| Maximum multiclass Macro F1 standard deviation | 0.10               |
| Minimum validation fold requirement            | 15 completed folds |

## 7 Stage III - Best\_Model\_Test.py

### 7.1 Test-eligible model selection

The model-confirmation stage reads eligible candidates from each target's validation results. Rank-one selection chooses the highest-ranked eligible configuration. An explicit selection mapping can instead identify a saved rank or an exact model/parameter combination. The selection is made before the model-confirmation block is scored.

### 7.2 Final chronological fit and test

The selected estimator is fitted using the preceding train and validation observations, with the target purge applied at the confirmation boundary. It then generates predictions for the 2013-2015 model-confirmation block. Each target is evaluated using the metrics appropriate to its statistical type.

### 7.3 Stored model-test metrics

| Type       | Stored metrics                        | Portfolio-retention setting         |
|------------|---------------------------------------|-------------------------------------|
| Continuous | RMSE, MAE, R <sup>2</sup> , Rank IC   | Minimum Rank IC: 0.10               |
| Binary     | ROC AUC, PR AUC, log loss, F1         | Minimum ROC AUC: 0.60; PR AUC: 0.20 |
| Multiclass | Macro F1, balanced accuracy, log loss | Minimum Macro F1: 0.45              |

Final\_Test\_Results.db stores the selected model, its serialised parameters, target identity and model-test metrics. The retained portfolio-facing rows also carry the metadata required for horizon and portfolio research.

## 8 From model test to portfolio metadata

### 8.1 Predictability Score

Predictability Score supplies a compact statistical summary for each retained target. Its definition follows the target's statistical type.

$$\text{Continuous: } P = \max(\text{Rank IC}, 0)$$

$$\text{Binary: } P = 0.50 [2 \max(\text{ROC AUC} - 0.50, 0)] + 0.30 \text{ PR AUC} + 0.20 F1$$

$$\text{Multiclass: } P = \text{Macro F1}$$

### 8.2 Portfolio Target Type

Each target is mapped to an economic type. The fine-grained classification preserves distinctions between absolute and relative return, directional events, volatility, downside behaviour, tail events, excursions and special market characteristics.

| Broad group | Example fine-grained types                                                 |
|-------------|----------------------------------------------------------------------------|
| Ranking     | ALPHA, RELATIVE_ALPHA, RISK_ADJUSTED_ALPHA, CROSS_SECTION_ALPHA            |
| Direction   | DIRECTION, DIRECTION_MULTICLASS, ALPHA_BINARY, BARRIER_ALPHA               |
| Risk        | VOLATILITY, DOWNSIDE_VOLATILITY, DOWNSIDE, TAIL_RISK, TAIL_EVENT           |
| Opportunity | ABSOLUTE_MOVE, UPSIDE_VOLATILITY, UPSIDE_EVENT, RECOVERY, REVERSAL         |
| Special     | Excursion timing, EXECUTION, LIQUIDITY, MARKET_IMPACT, CORRELATION, REGIME |

### 8.3 Horizon extraction

The prediction horizon is stored explicitly alongside the target. Daily portfolio research uses canonical horizon keys such as 1d, 5d, 20d, 60d, 120d and 252d. This metadata identifies both the horizon-score parameter and the signal-refresh interval.

### 8.4 Quality Score

Absolute quality measures normalise the selected model's statistical evidence. Within each Portfolio Target Type, the relative Quality Score places the retained targets on a common scale. A single target, or a group with equal absolute quality, receives a neutral relative score of 0.5. The resulting quality information is carried into the cached signal construction.

## 9 Stage IV - Horizon Score Backtests.py

### 9.1 Research-period role

Horizon research evaluates the relative influence of target/horizon components over 2016-2018. Selected target models are fitted on earlier research observations with the appropriate boundary purge. Their later predictions form the common input for the horizon-weight comparisons.

### 9.2 Predict once, vary horizon scores

The stage prepares predictions for each retained target and combines them into one research panel. A horizon configuration changes how those predictions contribute to the portfolio. The underlying selected model predictions can therefore be reused across many complete weighting experiments.

### 9.3 Horizon-score catalogue

The daily extensive profile varies six predefined target/horizon components. Each has three candidate weights. Other represented components are held at a weight of 1.0.

| Portfolio Target Type | Horizon | Candidate weights |
|-----------------------|---------|-------------------|
| ALPHA                 | 20d     | 0.0, 0.5, 1.0     |
| ALPHA                 | 60d     | 0.0, 0.5, 1.0     |
| RELATIVE_ALPHA        | 20d     | 0.0, 0.5, 1.0     |
| RELATIVE_ALPHA        | 60d     | 0.0, 0.5, 1.0     |
| DIRECTION             | 20d     | 0.0, 0.5, 1.0     |
| DOWNSIDE              | 20d     | 0.0, 0.5, 1.0     |

$$\text{Maximum declared horizon-weight combinations} = 3^6 = 729$$

Only target/horizon pairs represented by the retained model signals enter the active search. A zero candidate weight switches off that component for the comparison; positive candidates control its contribution relative to the other components.

### 9.4 Economic interpretation

Horizon research asks whether a portfolio benefits from emphasising short-, medium- or longer-horizon information within the selected target types. The retained configurations become reusable inputs for the later portfolio-setting sweep.

## 9.5 Portfolio backgrounds during horizon research

The Cartesian background design compares horizon configurations under three broad signal-group configurations, three stock-weight caps and three concentration settings. The same backgrounds are used to make the candidate comparisons.

| Configuration         | Ranking | Direction | Risk | Opportunity | Special |
|-----------------------|---------|-----------|------|-------------|---------|
| Equal Weight Baseline | .20     | .20       | .20  | .20         | .20     |
| Ranking Focused       | .50     | .20       | .20  | .10         | .00     |
| Risk Aware            | .35     | .20       | .35  | .10         | .00     |

  

| Dimension                 | Values                     |
|---------------------------|----------------------------|
| Maximum stock weight      | 0.10, 0.05, 0.15           |
| Concentration penalty     | 0.25, 0.00, 0.50           |
| Signal-refresh multiplier | 1.0                        |
| Background count          | $3 \times 3 \times 3 = 27$ |

## 9.6 Benchmark-relative Backtest Quality

Backtest Quality combines relative Sharpe, total return, maximum drawdown and average drawdown. Return and Sharpe improvements receive positive scores; drawdown components reward smaller drawdown magnitudes. Each component is scaled by the combined magnitude of the strategy and benchmark quantities.

$$BQ = 0.25 q_{Sharpe} + 0.35 q_{Return} + 0.25 q_{Max\ DD} + 0.15 q_{Average\ DD}$$

The benchmark and strategy use the same evaluation dates. This gives horizon research one combined portfolio criterion while retaining the underlying return and risk measures for inspection.

## 9.7 Sensitivity analysis and optional random screens

Sensitivity analysis examines the change in portfolio quality as one horizon component varies. The research settings include a 0.003 quality-range freeze threshold and a 0.002 near-best tolerance. Random-context rounds are also available through finite, explicitly configured schedules. The declared extensive profile uses the bounded candidate grid with those random rounds left empty.

## 9.8 Exhaustive survivor search

The remaining candidate lists are combined and evaluated exhaustively once they lie within the configured maximum of 1,000 combinations. With all six declared variable components represented, the initial grid contains 729 combinations. Each combination can be assessed against all 27 portfolio backgrounds.

The retention process preserves strong horizon configurations and compares their similarity. The configuration-correlation threshold is 0.95. The retained mappings are written to `Top_Horizon_Scores.txt` in a fixed order, which later stages reference through the Horizon Score Index.

## 10 Stage V - Backtest Database.py

### 10.1 Purpose

The cache stage converts selected model specifications into a reusable prediction database for strategy research. It reads `Final_Test_Results.db`, the selected-feature mapping and the configured stock universe, then rebuilds the required feature/target inputs and fits the selected models using the specified earlier training period.

### 10.2 Research windows

| Cache phase         | Training end     | Prediction period                  |
|---------------------|------------------|------------------------------------|
| Portfolio selection | 31 December 2018 | 1 January 2019 to 30 December 2022 |
| Final evaluation    | 31 December 2022 | 1 January 2023 to 31 August 2026   |

CACHE\_PHASE selects the corresponding settings in the root script. The selection and final caches are stored in separate phase directories. Their period-specific downloads provide the input observations needed by the selected features and models.

### 10.3 Feature history and benchmark

The configured download begins in 1997, supplying the historical observations used to build the selected rolling features. The required feature families are applied to the downloaded stock panel and market series. Model fitting and prediction then use the explicit training and backtest boundaries.

The benchmark is cached with the strategy research inputs so later portfolio evaluations can use a consistent market reference without repeating the preparation stage.

## 10.4 Cached database schema

Backtest\_Database.db provides a common interface to the portfolio stages. The strategy-facing Stocks table is a long-form signal panel. The Market table provides the corresponding benchmark information. Portfolio selection subsequently writes its Passed Strategies table to the selection database.

| Field group           | Role                                                                       |
|-----------------------|----------------------------------------------------------------------------|
| Date and Ticker       | Identify each observation and security.                                    |
| Return                | Realised return used by the later portfolio calculation.                   |
| Portfolio Target Type | Identify the economic signal category.                                     |
| Horizon Key           | Connect the signal to the horizon-score mapping.                           |
| Signal                | Store the prepared target-type/horizon signal used by the portfolio layer. |
| Passed Strategies     | Store selected Simulation IDs and their exact portfolio settings.          |

## 11 Stage VI - Backtest Simulations.py

### 11.1 Fine-grained portfolio signal structure

The portfolio stage joins each cached signal to its Horizon Score and Type Score. The Type Score is supplied by the broad group configuration: Ranking, Direction, Risk, Opportunity or Special. Each signal contributes according to both its horizon and its group weighting.

$$Contribution_{i,j,t} = Signal_{i,j,t} \times HorizonScore_j \times TypeScore_j$$

The refresh rule holds each target/horizon contribution for its configured number of observations before accepting the next signal value. Contributions are aggregated by stock and date. Direction information supplies an additional long-only gate: a negative aggregate direction makes the resulting stock score non-positive.

### 11.2 Universe comparisons

The selected stock universe supplies the principal portfolio test. Additional stock-removal and comparison-universe analyses investigate whether the strategy's result is broadly supported across its security inputs. The same signal and allocation rules are used when making those comparisons.

### 11.3 Portfolio allocation

Only positive stock scores enter the long-only allocation. Scores are scaled and normalised, then blended with an equal-weight component using the concentration setting. A capped redistribution assigns the available capital while respecting the maximum weight per stock.

$$u_i = (1 - \lambda) s_i / \sum s_j + \lambda / n$$

Here  $s$  denotes a positive stock score,  $n$  is the number of eligible stocks and  $\lambda$  is the concentration setting. A value of zero gives score-proportional starting weights; a value of one gives equal starting weights. The cap is applied through successive redistribution of the remaining allocation.

$$0 \leq w_i \leq w_{max}, \quad \sum w_i \leq 1$$

When the eligible positions do not use the full capital allocation, the remainder is held as cash. Each portfolio result records the stock weights used to earn the next observation's returns.

### 11.4 Portfolio-setting sweep

| Dimension                  | Research values                                    |
|----------------------------|----------------------------------------------------|
| Signal-group configuration | Equal Weight Baseline; Ranking Focused; Risk Aware |
| Refresh multiplier         | 0.5, 1.0                                           |
| Concentration penalty      | 0.00, 0.25, 0.50                                   |
| Maximum stock weight       | 0.05, 0.10                                         |
| Horizon mapping            | Each retained configuration from horizon research  |
| Robustness perturbations   | 100 configured iterations                          |

### 11.5 Position timing and P&L accounting

Scores observed at date  $t$  determine the allocation that earns the return at the next observation. Portfolio return is the sum of held stock weights multiplied by those stock returns, less any configured traded-notional cost. Changes in target weights provide the buy-plus-sell notional measure used by the cost calculation.

## 11.6 Stored experiment fields and summaries

Each simulation is identified by a Simulation ID. Its stored settings include the horizon configuration index, signal-group configuration, refresh multiplier, maximum weight and concentration penalty. Result fields describe return, Sharpe, drawdown, benchmark-relative quality and the applicable robustness measurements.

Portfolio\_Simulation\_Results.db contains the experiment results used to compare candidates and benchmark behaviour. The selected configurations are carried forward with their full setting identity so that the precise evaluation can reconstruct the same allocation rule.

## 12 Selecting simulations for precise testing

The selection process seeks candidates supported by several forms of portfolio evidence. It examines benchmark-relative performance, target-role behaviour, parameter stability and transfer across stock inputs. Similarity checks compare return streams, relative performance and stock overlap so the retained set can represent distinct portfolio behaviour.

| Similarity dimension                      | Configured value |
|-------------------------------------------|------------------|
| Return correlation threshold              | 0.95             |
| Relative performance difference threshold | 0.10             |
| Average stock-overlap threshold           | 0.80             |

Robustness analysis considers how quality changes when settings or stock inputs vary. The selected row retains the exact settings associated with its result; the subsequent stage uses those settings rather than reconstructing a strategy from a descriptive label.

### 12.1 Evidence carried into detailed evaluation

- A complete identity for the horizon and portfolio configuration.
- The original strategy and benchmark metrics used in comparison.
- The relevant portfolio-role and neighbourhood measurements.
- Daily data sufficient to reconstruct the chosen strategy.
- A record of the selected candidates entering the next research layer.

The intended output is a research shortlist whose behaviour can be examined in detail, followed by the account-level acceptance check and a frozen final-period evaluation.

## 13 Stage VII - Precise Backtest.py

### 13.1 Exact chosen configurations

Precise Backtest.py reconstructs the selected portfolio configurations from the cached signals and their saved settings. It checks the resulting metrics against the saved simulation quantities and produces a detailed record for each candidate.

The exact configuration includes the Horizon Score Index, Type Configuration, Rebalance Multiplier, Max Weight and Concentration Penalty. The script uses the same shared signal transformation and portfolio allocation functions as the simulation layer.

### 13.2 Daily return and allocation evidence

The detailed evaluation stores daily portfolio results, stock weights and traded-notional measurements. The daily stream supports return-distribution analysis, cost scenarios, rolling quality calculations and comparisons among finalists.

| Diagnostic group                | Research question                                                            |
|---------------------------------|------------------------------------------------------------------------------|
| Reconstructed summary           | Does the exact saved setting reproduce its recorded portfolio behaviour?     |
| Daily returns and stock weights | Which realised path and allocation generated the result?                     |
| Best-period removal             | How does the result behave when particularly favourable periods are removed? |
| Rolling 252-date quality        | How does a one-year section compare with the broader evaluation?             |
| Stock-removal tests             | How does quality change when individual security inputs are removed?         |
| Neighbouring settings           | How stable is performance near the chosen configuration?                     |
| Cost scenarios                  | How does traded notional translate into net-return sensitivity?              |

### 13.3 Retaining exact research identity

The detailed output associates each diagnostic with its Simulation ID and saved row. Daily streams, neighbours, stock removals and other intermediate results can therefore be traced back to the same candidate. These records support subsequent analysis without requiring the researcher to infer which settings produced a chart or summary.

### 13.4 Robustness and distribution diagnostics

The precise stage examines the strategy through several complementary transformations of its return path. It removes strong days, weeks, months and years; evaluates rolling annual windows; and measures how the result responds to the removal of individual stocks.

Neighbourhood evaluation perturbs the refresh multiplier, concentration setting and maximum weight around the chosen configuration. The results describe both the change in portfolio quality and the proportion of neighbouring runs retaining positive compounded returns after the configured execution allowance.

The return distribution is summarised using annualised Sharpe and expected shortfall. Probabilistic Sharpe and Deflated Sharpe calculations provide additional research statistics, with their calibration inputs saved alongside the results.

### 13.5 Research settings

| Setting                     | Value / role                                                      |
|-----------------------------|-------------------------------------------------------------------|
| Annualisation               | 252 daily observations                                            |
| Random seed                 | 42                                                                |
| DSR calibration sample size | Up to 1,000 sampled configurations from the saved grid            |
| DSR trial count             | Saved unique grid count, or an explicitly supplied research count |
| Execution-cost fraction     | 0.0005 of buy-plus-sell target-weight notional                    |
| Reference annual rate       | 0.0                                                               |
| Rolling evaluation window   | 252 observations                                                  |

### 13.6 Detailed output

The stage produces Final\_Strategy\_Evaluation.db, a CSV summary, daily-return and correlation CSV files, DSR calibration results and detailed per-strategy records. Evaluation\_Assumptions.json accompanies the output with the selected metric, cost and calibration conventions.

These outputs provide the analytical view of each selected portfolio before it moves to the GBP account check. They preserve both the summary metrics and the supporting return paths required to interpret them.

## 14 Account evaluation and completion of coded research

### 14.1 GBP Portfolio Check.py

The account stage evaluates selected strategies in the currency of the research portfolio. The initial capital is £4,000. Individual US-stock trades settle from and into a retained USD balance; the entire USD sleeve, including idle USD cash, is valued in GBP using the phase-specific exchange-rate series.

GBP\_per\_USD.csv supplies the GBP value of one USD for each required observation date. The preparation stage downloads the GBP/USD close series and converts its quotation direction. The account calculation joins exchange rates to the strategy dates before constructing GBP returns and equity.

$$1 + r_{GBP,t} = (1 + r_{USD,t}) \times x_t / x_{t-1}$$

Here x is GBP per USD. The initial currency conversion and the configured terminal conversion each use a 0.15% conversion fee. Individual USD trades use zero broker commission. A separate execution allowance of 0.05% of traded notional represents the selected spread, slippage and trading-charge research scenario.

### 14.2 Drawdown acceptance

The GBP equity path begins from the initial capital and is compared with its running peak. A candidate passes the configured account criterion when its maximum GBP drawdown is no greater than 20%.

$$Drawdown_t = Equity_t / \max(Equity_0, \dots, Equity_t) - 1$$

The account database retains the evaluated candidates and their daily paths. A separate Passed Strategies table carries the candidates satisfying the GBP drawdown criterion into the final evaluation phase.

| Account setting        | Research specification                                          |
|------------------------|-----------------------------------------------------------------|
| Initial capital        | £4,000                                                          |
| Account and settlement | Trading 212 Invest; individual trades; retained USD             |
| Currency conversion    | 0.15% on entering USD and on the configured terminal conversion |
| GBP drawdown threshold | 20% from the running equity peak                                |

### 14.3 Frozen Final Test.py

The final stage evaluates the retained strategy settings against the separate final-period cache. Model and portfolio choices are completed before this stage begins. The final cache uses training observations through 2022 and supplies the configured January 2023-August 2026 evaluation period.

Each finalist is reconstructed with its saved horizon mapping, signal-group configuration, refresh multiplier, maximum weight and concentration setting. The GBP account settings remain consistent with the earlier account-selection check.

- The selection database identifies every finalist and its exact settings.
- The later cache supplies the final-period stock signals and returns.
- The frozen horizon file supplies the configuration mapping and order.
- The final-phase FX file supports GBP valuation.
- The final output reports every selected strategy and its realised evaluation metrics.

### 14.4 Final output and research record

Frozen\_Final\_Evaluation.db contains Daily Returns, Summary and Frozen Settings tables. The summary reports total return, annualised volatility, Sharpe and drawdowns, together with the GBP drawdown-threshold outcome. A companion CSV makes the summary available for further analysis.

The final manifest records the input-file identities, declared and actual evaluation dates, account settings, annualisation and signal-group configuration. The output also includes a database-check contract for independent metric reconstruction from the saved daily stream.

### 14.5 Intended use of the completed research

The completed sequence provides a documented set of candidate strategies and the evidence supporting their selection. Its intended use is to support stock-market research, compare portfolio allocation rules, examine their behaviour across the designated periods and prepare a clearly specified strategy for prospective observation.

The researcher can trace a final candidate back through its GBP account evaluation, precise diagnostics, portfolio settings, horizon configuration, selected models, validation history and original feature mapping. This traceable research chain is the central project output.

## 15 Storage architecture

| Store                                      | Purpose                                                                                 |
|--------------------------------------------|-----------------------------------------------------------------------------------------|
| Features_Targets_Data.db                   | Stock-universe tables containing the screened feature and target panel.                 |
| Selected_Features.txt                      | Universe-keyed mappings from target names to selected feature names.                    |
| Validation_Model_Fits/High_Liquidity_30.db | Target-specific fold histories and search summaries.                                    |
| Final_Test_Results.db                      | Model-confirmation metrics and portfolio-facing target metadata.                        |
| Top_Horizon_Scores.txt                     | Ordered retained horizon-score mappings.                                                |
| Backtest_Database.db                       | Cached strategy and market inputs, with selected portfolio rows in the selection phase. |
| Portfolio_Simulation_Results.db            | Portfolio experiments, comparison metrics and robustness results.                       |
| Final_Strategy_Evaluation.db               | Detailed precise-evaluation summaries and associated research outputs.                  |
| GBP Check/GBP_Selection.db                 | GBP account summaries, daily results and the account-approved selection.                |
| Frozen_Final_Evaluation.db                 | Final-period daily returns, summary metrics and frozen settings.                        |

### 15.1 Phase directories

The configured directory root is `data/extensive_20260904`. Its `model_research`, `portfolio_selection` and `final_evaluation` subdirectories separate the three phases. Each script uses its explicit input/output paths. The preparation stage creates these locations and copies the completed artifacts needed by the next phase.

### 15.2 Database checking

`Check Databases.py` reads stored results and checks their structure and internal consistency. Applicable checks cover table schemas, unique identifiers, parameter dictionaries, date boundaries, fold summaries, related simulation identities and the reconstruction of summary metrics from daily return streams.

The account and final evaluators generate a metric-replay contract with their output. The research record therefore includes both the stored results and a repeatable way to check the calculations represented by those results.

## 16 Intended research sequence

| Phase              | Sequence                                                                                                                                           |
|--------------------|----------------------------------------------------------------------------------------------------------------------------------------------------|
| Model research     | Prepare directories; create and screen the data; fit and validate models; confirm selected models; select horizon configurations.                  |
| Portfolio research | Prepare selection inputs and FX history; build the selection cache; run simulations; perform precise diagnostics; apply the GBP account criterion. |
| Final evaluation   | Prepare frozen final inputs and FX history; build the final cache; evaluate all frozen finalists.                                                  |

Run `Research.py` provides a single launcher for one step at a time. Its stage setting selects the script. `Prepare Research.py` changes phase when the research moves from models to portfolio selection and then to final evaluation. `Backtest Database.py` uses `CACHE_PHASE` to choose the selection or final prediction window.

## 17 What counts as evidence at each layer

| Layer              | Question                                               | Main evidence                                                         |
|--------------------|--------------------------------------------------------|-----------------------------------------------------------------------|
| Data screening     | What information enters the research?                  | Training-period coverage, validity, continuity and retained features. |
| Model validation   | Which configurations remain competitive?               | Fold metrics, stability, pruning decisions and eligibility.           |
| Model confirmation | How does the selected model perform later?             | Target-specific confirmation metrics and model metadata.              |
| Horizon research   | How should prediction horizons contribute?             | Matched background tests and retained horizon mappings.               |
| Portfolio research | Which allocation rules deserve detailed study?         | Benchmark comparison, robustness and similarity results.              |
| Precise evaluation | What supports the selected portfolio result?           | Reconstructed streams, perturbations, removals and cost scenarios.    |
| GBP account check  | Does the account meet the chosen drawdown criterion?   | GBP equity, net returns and peak-to-trough drawdown.                  |
| Final evaluation   | How do the frozen choices perform in the final period? | All finalist outcomes, daily returns and the saved research manifest. |

Together these layers connect mathematical definitions to model evidence, portfolio decisions and account-level results. The structure supports repeated research with explicit settings and a complete record of the choices being evaluated.

## A Key formula summary

### A.1 Predictability Score

$$P_{continuous} = \max(IC, 0)$$

$$P_{binary} = 0.50 [2 \max(AUC - 0.50, 0)] + 0.30 PR AUC + 0.20 F1$$

$$P_{multiclass} = Macro F1$$

### A.2 Relative Quality Score

For n retained targets of one Portfolio Target Type, let a be the absolute quality measure and define the lower bound  $l = 0.5/n$  and upper bound  $u = 1 - l$ .

$$Q_j = l + [(a_j - a_{min}) / (a_{max} - a_{min})] (u - l)$$

A one-target group, or a group with equal absolute-quality values, uses  $Q = 0.5$ . Cached signal preparation carries the model-quality information into the portfolio-facing signal.

### A.3 Horizon and type contribution

$$C_{i,j,t} = S_{i,j,t} H_j T_j$$

S is the cached signal, H the horizon score and T the score of the signal's broad portfolio group. Contributions are combined by stock and date, with the direction gate applied before long-only allocation.

### A.4 Horizon refresh

$$Refresh\ interval = \max(1, \text{ceiling}(m \times h))$$

The multiplier m acts on the target horizon h. The contribution is held for the resulting number of observed rows within its stock/type/horizon series before the next refresh.

### A.5 Horizon Backtest Quality

$$q_{return} = (R_s - R_b) / (|R_s| + |R_b| + \epsilon)$$

$$q_{drawdown} = (|D_b| - |D_s|) / (|D_b| + |D_s| + \epsilon)$$

The Sharpe component follows the return form. Maximum and average drawdown use the drawdown form. The horizon criterion weights Sharpe, return, maximum drawdown and average drawdown by 0.25, 0.35, 0.25 and 0.15 respectively.

## A.6 Portfolio weights

$$u_i = (1 - \lambda) s_i / \sum s_j + \lambda / n$$

Positive scores define the eligible stock set. The allocation uses the score/equal-weight blend, then caps and redistributes the remaining capital subject to the maximum stock weight.

## A.7 Portfolio return and traded notional

$$Turnover_t = \sum |w_{i,t} - w_{i,t-1}|$$

$$r_{p,t+1} = \sum w_{i,t} r_{i,t+1} - c \times Turnover_t$$

The initial stock entry is included in the traded-notional calculation. The cost fraction  $c$  is applied to total buy-plus-sell target-weight notional.

## A.8 Compounded return, volatility and Sharpe

$$Total\ return = \Pi (1 + r_p) - 1$$

$$Annualised\ volatility = sample\ standard\ deviation(r) \times \sqrt{252}$$

$$Sharpe = mean(r) / sample\ standard\ deviation(r) \times \sqrt{252}$$

## A.9 GBP valuation and drawdown

$$1 + r_{GBP,t} = (1 + r_{USD,t}) x_t / x_{t-1}$$

$$Equity_t = £4,000 \times \Pi (1 + r_{GBP,t})$$

The first and configured final return factors include the corresponding currency-conversion factor,  $1 - 0.0015$ . Maximum drawdown is the minimum of the equity-to-running-peak ratio minus one.

## A.10 Signal-group configurations

| Configuration         | Ranking | Direction | Risk | Opportunity | Special |
|-----------------------|---------|-----------|------|-------------|---------|
| Equal Weight Baseline | .20     | .20       | .20  | .20         | .20     |
| Ranking Focused       | .50     | .20       | .20  | .10         | .00     |
| Risk Aware            | .35     | .20       | .35  | .10         | .00     |

## B Horizon configurations and portfolio sweep

A retained horizon configuration is a complete mapping from Portfolio Target Type to horizon key and score. The extensive daily design declares six variable components, each using 0.0, 0.5 and 1.0, with other components fixed at 1.0. The ordered mappings are passed to the portfolio sweep.

| Dimension                          | Values                                                     |
|------------------------------------|------------------------------------------------------------|
| Horizon configurations             | Up to 729 in the declared variable grid                    |
| Horizon backgrounds                | 27 combinations of groups, caps and concentration settings |
| Portfolio weight caps              | 0.05; 0.10                                                 |
| Portfolio concentration            | 0.00; 0.25; 0.50                                           |
| Portfolio refresh multipliers      | 0.5; 1.0                                                   |
| Simulation perturbation iterations | 100                                                        |

## C Base universe definitions

### C.1 High Liquidity 30

AAPL, MSFT, AMZN, NVDA, META, TSLA, AMD, INTC, QCOM, MU, CSCO, ORCL, JPM, BAC, WFC, C, XOM, CVX, F, GM, T, VZ, PFE, JNJ, WMT, DIS, GE, HD, NFLX, GOOG

### C.2 Medium Liquidity 30

ADI, MCHP, TXN, STX, WDC, PNC, USB, BK, STT, COF, CAT, DE, EMR, ETN, ITW, LOW, TGT, KR, BBY, DRI, AMGN, GILD, BIIB, BMY, CVS, OXY, EOG, SLB, HAL, VLO

### C.3 Lower Liquidity 30

AIT, ARCB, BRC, CALM, CHCO, CPK, CNMD, FFIN, GATX, GBCI, GFF, HNI, HVT, JJSF, LANC, MGEE, MLAB, MMSI, MTRN, NRIM, NWN, RCKY, RELL, SCL, TNC, UVSP, WASH, WDFC, NATH, RES

### C.4 Sector Spread 30

MSFT, ADI, RELL, JPM, PNC, CHCO, GE, CAT, AIT, AMZN, LOW, HVT, WMT, KR, JJSF, JNJ, AMGN, CNMD, XOM, OXY, RES, NEE, DUK, CPK, FCX, NUE, SCL, PLD, SPG, UHT

### C.5 Liquidity Barbell 30

AAPL, MSFT, AMZN, NVDA, META, TSLA, JPM, BAC, XOM, CVX, WMT, JNJ, GE, HD, NFLX, AIT, ARCB, BRC, CALM, CHCO, CPK, CNMD, FFIN, GATX, HNI, JJSF, MGEE, MTRN, TNC, WDFC

## D Research settings reference

| Setting                            | Intended research value                                                  |
|------------------------------------|--------------------------------------------------------------------------|
| Initial model training             | 2000-2006                                                                |
| Walk-forward validation            | 2007-2012                                                                |
| Model confirmation                 | 2013-2015                                                                |
| Horizon selection                  | 2016-2018                                                                |
| Portfolio selection                | 2019-2022                                                                |
| Final historical evaluation        | January 2023-August 2026                                                 |
| Universe                           | High Liquidity 30                                                        |
| Validation window                  | 63 observed dates                                                        |
| Minimum validation folds           | 15                                                                       |
| Pruning folds                      | 3, 5, 7, 9, 11, 14                                                       |
| Pruning keep fractions             | 0.95, 0.90, 0.80, 0.65, 0.50, 0.50                                       |
| Pruning configuration caps         | 20,000; 5,000; 1,000; 300; 70; 15                                        |
| Model catalogue                    | Grid                                                                     |
| Model stability thresholds         | Rank IC SD 0.10; ROC AUC SD 0.07; Macro F1 SD 0.10                       |
| Horizon analysis mode              | Daily; Cartesian portfolio backgrounds                                   |
| Maximum exhaustive combinations    | 1,000                                                                    |
| Horizon freeze / near-best range   | 0.003 / 0.002                                                            |
| Horizon configuration correlation  | 0.95                                                                     |
| Portfolio neighbourhood iterations | 100                                                                      |
| Portfolio similarity checks        | Return correlation 0.95; performance difference 0.10; stock overlap 0.80 |
| Account capital and base currency  | £4,000; GBP                                                              |
| Account conversion fee             | 0.15% at each configured conversion                                      |
| Broker trading commission          | 0.00 for the retained-USD individual-trade profile                       |
| Execution-cost research scenario   | 0.05% of traded notional                                                 |
| GBP drawdown acceptance            | 20%                                                                      |
| DSR calibration sample             | Up to 1,000 configurations                                               |
| Run Research numerical threads     | 2 requested threads; 2 requested joblib CPUs                             |

The root scripts hold the editable settings for each stage. This reference records the intended research configuration, including the selected progressive-pruning schedule and 15-fold eligibility minimum.